Structure		External diameter without wrapping		External diameter with wrapping 1 x 52 (1 x Natural silk)		2 x 52 (2 x Natural silk)		Cross-section	DC-Resistance at 20°C (68°F)			Metres per kg
No. of strands	diameter of single wire [mm]	min [mm]	max [mm]	min [mm]	max [mm]	min [mm]	max [mm]	[mm²]	min [Ω/km]	nom [Ω/km]	max [Ω/km]	nom [m/kg]
	0.016 AWG 54	Other dimensions on request										
	0.020 AWG 52	Other dimensions on request										
	0.022 AWG 51	Other dimensions on request										
	0.025 AWG 50	Other dimensions on request										
10		0.127	0.142	0.157	0.177	0.187	0.212	0.00707	2176	2467	2713	11764
20		0.179	0.200	0.209	0.235	0.239	0.270	0.01414	1088	1233	1357	6060
25								0.01767				4964
30								0.02121				3921
35				External diameter depending on type				0.02474				3355
45				of wrapping. Dimensions on request				0.03181				2611
60	0.03							0.04241				1960
75								0.05301				1567
90	AWG 48							0.06362				1307
105								0.07422				1119
120								0.08482				980
135								0.09542				871
180								0.12723				653
225								0.15904				522
270								0.19085				435
Other types of litz wire up to external diameter 11 mm on request												
10		0.164	0.186	0.194	0.221	0.224	0.256	0.01282	1228	1387	1522	7847
12		0.183	0.208	0.213	0.243	0.243	0.278	0.1538	1023	1156	1268	6536
15		0.201	0.229	0.236	0.269	0.261	0.299	0.01923	819	925	1015	5229
20		0.232	0.264	0.267	0.304	0.292	0.334	0.02564	614	694	761	3922
25		0.260	0.295	0.295	0.335	0.320	0.365	0.03204	491	555	609	3137
30		0.284	0.323	0.319	0.363	0.344	0.393	0.03845	409	462	522	2615
35		0.307	0.349	0.342	0.389	0.367	0.419	0.04486	351	396	448	2241
45	0.04	0.348	0.395	0.383	0.435	0.408	0.465	0.05768	273	308	348	1743
60		0.405	0.460	0.440	0.500	0.465	0.530	0.07691	205	231	266	1231
75	AWG 46	0.453	0.515	0.488	0.555	0.513	0.585	0.09613	164	185	213	985
90		0.497	0.565	0.532	0.605	0.567	0.645	0.11536	136	154	178	821
105		0.537	0.610	0.572	0.650	0.607	0.690	0.13459	117	132	152	704
120		0.572	0.650	0.607	0.690	0.642	0.730	0.15381	102	116	133	615
135		0.607	0.690	0.642	0.730	0.677	0.770	0.17304	91	103	118	547
180		0.722	0.820	0.757	0.860	0.792	0.900	0.23072	68	77	90	385
225		0.805	0.915	0.840	0.955	0.895	1.015	0.28840	55	62	72	308
270		0.884	1.005	0.919	1.045	0.974	1.105	0.34608	45.5	51.4	60.3	257
Other types of litz wire up to external diameter 11 mm on request												
6		0.162	0.179	0.192	0.214	0.222	0.249	0.01202	1320	1480	1613	8505
8		0.186	0.206	0.216	0.241	0.246	0.276	0.01602	990	1100	1210	6379
10		0.209	0.231	0.244	0.271	0.269	0.301	0.02003	792	888	968	5103
12		0.232	0.257	0.267	0.297	0.292	0.327	0.02403	660	740	807	4253
15		0.256	0.283	0.291	0.323	0.316	0.353	0.03004	528	592	645	3402
20		0.295	0.327	0.330	0.367	0.355	0.397	0.04006	396	444	484	2552
25		0.330	0.366	0.365	0.406	0.390	0.436	0.05007	317	355	387	2041
30		0.382	0.401	0.397	0.441	0.422	0.471	0.06008	264	296	332	1701
35	0.05	0.391	0.433	0.426	0.473	0.451	0.503	0.07010	226	254	285	1458
45		0.443	0.490	0.478	0.530	0.503	0.560	0.09012	176	197	222	1134
60	AWG 44	0.515	0.570	0.550	0.610	0.585	0.650	0.12017	132	148	169	801
75		0.577	0.639	0.612	0.679	0.647	0.719	0.15021	106	118	136	641
90		0.633	0.701	0.668	0.741	0.703	0.781	0.18025	88	99	113	534
105		0.683	0.756	0.718	0.796	0.753	0.836	0.21029	75	85	97	458
120		0.728	0.806	0.763	0.846	0.798	0.886	0.24033	66	74	85	400
135		0.773	0.856	0.808	0.896	0.863	0.956	0.27037	59	66	75	356
180		0.918	1.017	0.953	1.057	1.008	1.117	0.36050	44.0	49.3	57.6	250
225		1.025	1.135	1.060	1.175	1.115	1.235	0.45062	35.2	39.5	46.0	200
270		1.126	1.246	1.161	1.286	1.216	1.346	0.54075	29.3	32.9	38.4	167
Other types of litz wire up to external diameter 11 mm on request												
3		0.163	0.189	0.193	0.224	0.223	0.259	0.01212	1331	1468	1578	8537
5		0.205	0.238	0.240	0.278	0.265	0.308	0.02019	799	881	947	5122
6		0.220	0.254	0.255	0.294	0.280	0.324	0.02423	666	734	789	4268
8		0.253	0.293	0.288	0.333	0.313	0.363	0.03231	499	551	592	3201
10		0.283	0.328	0.318	0.368	0.343	0.88	0.4038	399	440	473	2561
12		0.315	0.365	0.350	0.405	0.375	0.435	0.04846	333	367	394	2134
15		0.347	0.402	0.382	0.442	0.407	0.472	0.06058	266	294	316	1702
20		0.401	0.464	0.436	0.504	0.461	0.534	0.08077	200	220	237	1280
25		0.448	0.519	0.483	0.559	0.508	0.589	0.10096	160	176	189	1024
30		0.491	0.568	0.526	0.608	0.561	0.648	0.12115	133	147	163	854
35		0.530	0.614	0.565	0.654	0.600	0.694	0.14134	114	126	139	732
45		0.601	0.696	0.636	0.736	0.671	0.776	0.18173	89	98	108	570
60		0.699	0.810	0.734	0.850	0.769	0.890	0.24230	67	73	83	402
75		0.783	0.906	0.813	0.946	0.873	1.006	0.30288	53	59	66	322
90		0.859	0.994	0.894	1.034	0.949	1.094	0.36345	44.4	48.9	55.2	268
105		0.927	1.074	0.962	1.114	1.017	1.174	0.42403	38.0	41.9	47.3	230
120	0.071	0.988	1.144	1.023	1.184	1.078	1.244	0.48461	33.3	36.7	41.4	201
135		1.049	1.214	1.084	1.254	1.139	1.314	0.54518	29.6	32.6	36.8	179
180	AWG 41	1.246	1.443	1.281	1.483	1.336	1.543	0.72691	22.2	24.5	28.2	126
225		1.391	1.610	1.426	1.650	1.481	1.710	0.90864	17.8	19.6	22.5	107
270		1.528	1.769	1.563	1.809	1.618	1.869	1.09036	14.8	16.3	18.8	94
315		1.649	1.910	1.684	1.950	1.739	2.010	1.27209	12.7	14.0	16.1	72
405		1.870	2.165	1.905	2.205	1.960	2.265	1.63555	9.9	10.9	12.5	56
420			2.13					1.67	9.5	10.5	11.3	54
525			2.39	External diameter depending on type				2.08	7.6	8.4	9.0	43
630			2.61	of wrapping. Dimensions on request				2.50	6.3	7.0	7.5	36
735			2.82					2.92	5.4	6.0	6.4	31
840			3.03					3.33	4.8	5.2	5.6	27
945			3.20					3.75	4.3	4.7	5.0	24
1260			3.70					5.00	3.2	3.5	3.8	17
1575			4.15					6.21	2.5	2.8	3.0	13
1890			4.55					7.49	2.1	2.3	2.5	11
2205			4.86					8.74	1.8	2.0	2.1	10
2835			5.60					11.23	1.4	1.6	1.7	7.5
Other types of litz wire up to external diameter 11 mm on request												
10		0.407	0.451	0.442	0.491	0.467	0.521	0.08011	205	222	235	1300
12		0.452	0.502	0.487	0.542	0.512	0.572	0.09613	170	185	196	1079
15		0.498	0.553	0.533	0.593	0.568	0.633	0.12017	136	148	157	803
20		0.574	0.638	0.609	0.678	0.644	0.718	0.16022	102	111	118	647
25		0.642	0.714	0.678	0.754	0.713	0.794	0.20028	82	89	94	518
30		0.704	0.782	0.739	0.822	0.774	0.862	0.24033	68	74	81	432
35		0.761	0.845	0.796	0.885	0.851	0.945	0.28039	58	63	69	370
45		0.862	0.957	0.897	0.997	0.952	1.057	0.36050	45.5	49.3	53.9	288
60		1.003	1.113	1.038	1.153	1.093	1.213	0.48066	34.1	37.0	41.2	203
75		1.123	1.246	1.158	1.286	1.213	1.346	0.60083	27.3	26.9	33.0	163
90		1.232	1.367	1.267	1.407	1.322	1.467	0.72100	22.7	24.7	27.5	136
105		1.330	1.476	1.365	1.516	1.420	1.576	0.84116	19.5	21.1	23.5	116
120		1.417	1.573	1.452	1.613	1.507	1.673	0.96133	17.0	18.5	20.6	102
135	0.10	1.504	1.670	1.539	1.710	1.594	1.770	1.08150	15.2	16.4	18.3	90
140			1.65					1.10	14.6	15.9	16.8	87
175	AWG 38		1.83	External diameter depending on type				1.38	11.7	12.7	13.4	70
210			2.01	of wrapping. Dimensions on request				1.65	9.8	10.6	11.2	58
245			2.16					1.93	8.4	9.1	9.6	50
280			2.34					2.20	7.3	7.9	8.4	44
350			2.62					2.75	5.9	6.3	6.7	35
420			2.95					3.30	4.9	5.3	5.6	27
525			3.27					4.13	3.9	4.2	4.5	22
630			3.59					4.95	3.3	3.5	3.7	18
735			3.87					5.77	2.8	3.0	3.2	16
840			4.19					6.60	2.4	2.6	2.8	14
945			4.40					7.42	2.2	2.3	2.5	12
1050			4.68					8.25	1.9	2.1	2.2	11
1260			5.12					9.90	1.6	1.8	1.9	9
1400			5.49					10.99	1.5	1.6	1.7	8
Other types of litz wire up to external diameter 11 mm on request												
3			0.49	External diameter depending on type				0.10	178	185	188	1103
4			0.58	of wrapping. Dimensions on request				0.13	134	139	141	827
5			0.62					0.16	107	111	113	661
6			0.66					0.19	89	92	94	551
7			0.73					0.22	76	79	81	472
8			0.77					0.25	67	69	71	414
9			0.83					0.28	59	62	63	368
10			0.86					0.31	53	55	57	331
15			1.05					0.47	36	37	38	221
20			1.21					0.63	27	27.7	28.2	165
25			1.36					0.79	21	22.2	22.6	132
30			1.49					0.94	18	18.5	18.8	110
35			1.61					1.10	15	15.8	16.1	95
40			1.72					1.26	13	13.9	14.1	83
45			1.82					1.41	11.8	12.3	12.6	74
50			1.93					1.57	10.7	11	11.3	66
60	0.20		2.11					1.89	8.9	9.2	9.4	52
80			2.47					2.51	6.7	6.9	7.1	39
90	AWG 32		2.59					2.83	5.9	6.2	6.3	35
100			2.65					3.14	5.3	5.5	5.7	31
105			2.80					3.30	5.1	5.3	5.4	30
120			3.00					3.77	4.5	4.6	4.7	26
135			3.17					4.24	4.0	4.1	4.2	23
150			3.36					4.71	3.6	3.7	3.8	21
180			3.76					5.66	3.0	3.08	3.14	16
200			3.94					6.28	2.7	2.77	2.83	15
250			4.22					7.86	2.1	2.22	2.26	13
300			4.52					9.43	1.8	1.85	1.88	10
350			4.97					11.0	1.5	1.58	1.61	9
360			5.30					11.31				9.3
600			7.10					18.84				5
800			7.50					25.13				4
1000			9.00					31.42				3.3
1200			10.50					37.70				2.7
1400			11.00					43.98				2.3
Other types of litz wire up to cross-section 500 mm² on request												
3			0.85	External diameter depending on type				0.30	57.0	58.7	59.7	345
4			1.00	of wrapping. Dimensions on request				0.40	43.0	44.0	44.8	259
5			1.07					0.50	34.0	35.2	35.8	207
6			1.14					0.59	28.5	29.3	29.8	172
7			1.26					0.69	24.4	25.1	25.6	147
8			1.32					0.79	21.4	22.0	22.4	129
9			1.42					0.89	19.0	19.5	19.9	115
10			1.47					0.99	17.1	17.6	17.9	103
15	0.355		1.81					1.49	11.4	11.7	11.9	73
20			2.08					1.98	8.6	8.8	9.0	52
25	AWG 27		2.33					2.48	6.8	7.0	7.2	42
30			2.56					2.97	5.70	5.87	5.97	32
35			2.76					3.47	4.90	5.03	5.11	29
40			3.00					3.96	4.30	4.4	4.48	24
45			3.15					4.46	3.80	3.91	3.98	21
50			3.22					4.95	3.42	3.52	3.58	20
60			3.62					5.94	2.85	2.93	2.98	16
75			4.06					7.42	2.28	2.35	2.39	13
Other types of litz wire up to external diameter 11 mm on request												

Type of coil	Fig.:	d 1 [mm]	d 2 [mm]	d 3 [mm]	d 4 [mm]	L 1 [mm]	L 2 [mm]	Tare-weight [kg]	Filled-weight [kg]
K 63	1	63	40	11	–	63	49	0.040	0.2
K 80	1	80	50	16	–	80	64	0.070	0.7
K 100	1	100	53	16	–	100	80	0.125	1.4
K 125	1	125	80	16	–	125	100	0.200	2.8
HK 125/45	2	125	63	16	–	125	56	0.112	3.0
K 160	1	160	100	22	–	160	128	0.350	6.0
HKV 160/45	2	160	80	22	–	160	73	0.215	6.0
K 200	1	200	125	22	–	200	160	0.600	12.0
K 250	1	250	160	22	–	200	160	1.050	20.5
A 250/400	2	250	160	236	140	400	335	2.250	45.0
VB 250/400*)									
A 315/500	3	315	200	300	180	500	524	4.350	90.0
VB 315/500*)									
K 355	1	355	244	36	–	200	160	1.850	33.0
DFW 500	1	500	315	36	–	250	180	7.650	70.0
Other types of coil on request									

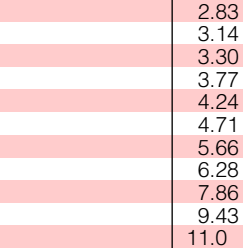


Fig. 1: Cylindrical coil

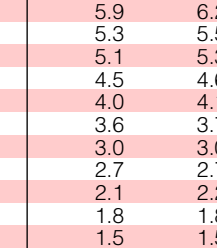


Fig. 2: Bi-conical coil

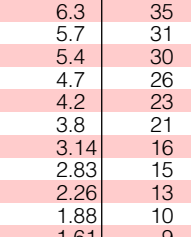


Fig. 3: Conical coil

Conforming to IEC 264-2-1 (DIN 46399) [Fig. 1+2]
IEC 264-3-1 (DIN 46383) [Fig. 3]

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Technical Data

RUPALIT® Litz wire

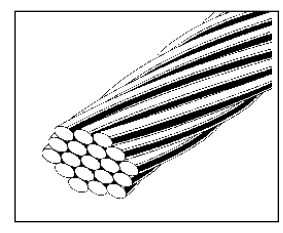
(The technical data comply in part with the specification given in standard IEC 60317-11)

RUPALIT® high-frequency litz wires are constructed from enamelled copper wire (magnet wire) made of RUPOL®, RUPEX® or RUTHERM® – see illustrations below. (Aluminium magnet wires on request)

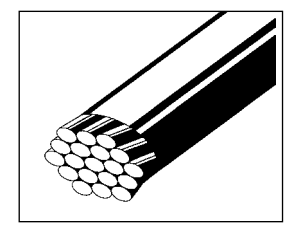
On special request, we can also supply any other design of litz wire up to an overall cross-section of 1.000 mm²

For further information, please refer to our Data Sheets

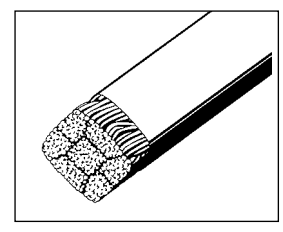
RUPALIT®



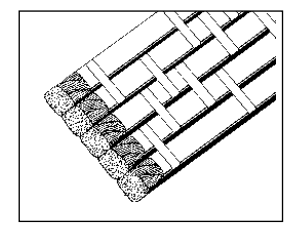
RUPALIT®



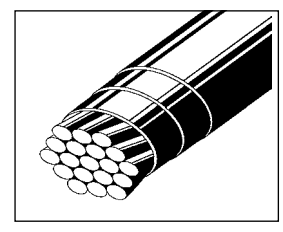
RUPALIT® Profile



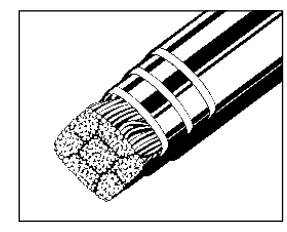
RUPALIT® Planar



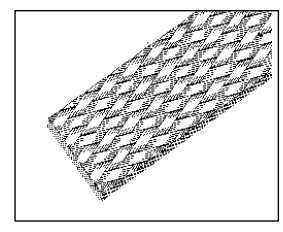
RUPALIT® Safety (IEC 60950)



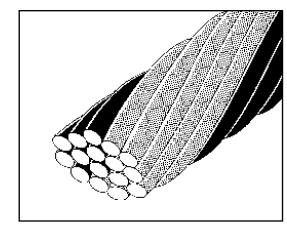
RUPALIT® Safety Profile (IEC 60950)



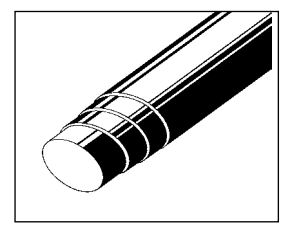
RUPALIT® Braided



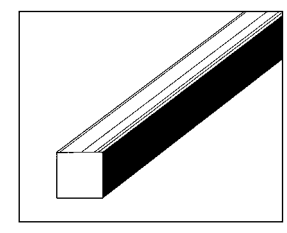
RUPALIT® 05



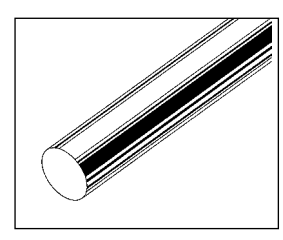
RUPOL® Safety (IEC 60950)



RUPA® Profile



RUPOL® · RUPEX® · RUTHERM®



RUPEX®

Magnet wire, heat-proof
IEC 60317-8; MW 74, 200°C
IEC 60317-13; MW 35, 220°C
IEC 60317-7; MW 16, 240°C

RUTHERM®

self bonding Magnet wire, solderable

IEC 60317-35; 155°C
IEC 60317-36; 180°C

RUPOL®

Magnet wire, solderable

IEC 60317-20; MW 79, 155°C
IEC 60317-52; MW 82, 180°C

RUPALIT® litz wire with UL approval

RUPALIT® litz wire with VDE approval confirming to IEC 60950

RUPALIT® Litz wire produced conforming to DIN EN ISO 9002

Braiding material with code number:

cotton yarn (50),
pure silk (52)*,
artificial silk (60)*,
acetate silk (61)*,
polyamide yarn (63)*,
arom. polyamide yarn (68) TI** 180°C (360°F),
ceramic yarn (69) TI** 1200°C (2200°F),
glass yarn impregnated (88) TI** 180°C (356°F)

Foil wrapping with code number:

polyester foil (25) TI** 150°C (302°F), polyamide foil (26) TI** 180°C (356°F),
polyimide foil (27) TI** 220°C (428°F),

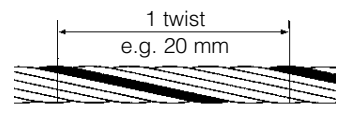
Sheathing with abbreviation:

PVC TI** 105°C (221°F), ETFE TI** 155°C, FEP TI** 155°C (311°F), Silicone (Si) TI** 180°C (356°F), Shrinkdown plastic tubing (SL) TI** 135°C (275°C), (SL) TI** 175°C (347°F), (SL) TI** 220°C (428°F)

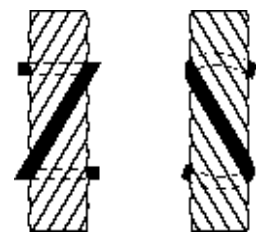
*UL-Approval for TI** 155°C

**TI: temp. limit index in °C (F°)

Length of twist:



Direction of twist:



Z clockwise

S anti-clockwise

PACK Quality Features

- RUPALIT® HF litz wire as focal point of our product range
- High quality standards
- Rapid delivery
- Special customer orders
- Flexible and speedy delivery, even for individual requirements
- In-house production of enamelled wire
- 100% quality check of single copper strands for RUPALIT®
- Specialized business with more than 70 years´ experience
- Distinguished, international cli entele in Europe and overseas
- Small quantities also available
- Production conforming to international standards (DIN, VDE, EN, IEC, UL), as well as special customer orders

Coils

The conical shape ot the coil core permits perfect over-end unwinding of extremely thin wires at high speeds, preventing tangling.

We can supply plastic coils made of highly impact-resistant polystyrene (PS) or acrylic nitrile-butadiene-styrene (ABS).

For coils returned to us in reusable condition, we will refund the purchase price minus 10% recycling costs.